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Applicant: SUMITOMO ELECTRIC INDUSTRIES, LTD.
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New claims

1. An optical connector (10) comprising:

an optical fiber (50) including a glass fiber (51, 51A, 51B, 510) and a resin coating covering the glass fiber (51, 51A, 51B, 510);

a front end portion;

a rear end portion positioned on a side opposite to the front end portion;

a ferrule assembly (100) including a ferrule (110) and a holding part, the ferrule (110) being fixed to, of the optical fiber (50), a tip portion of the glass fiber (51, 51A, 51B, 510) exposed from the resin coating, a rear end portion of the ferrule (110) being fixed to the holding part, the holding part being provided with a flange (130, 130A);

a housing (20, 30) including an inner wall surface, a reference surface (200A, 200B), and a positioning part (20A, 20B), the inner wall surface defining an internal space in which at least the holding part of the ferrule assembly (100) is accommodated, the reference surface (200A, 200B) constituting a part of the inner wall surface and being positioned to face a part of an outer peripheral surface of the flange (130, 130A), the positioning part (20A, 20B) being provided for restricting a movement of the holding part in the internal space; and

an elastic member (140) configured to urge the flange (130, 130A) toward the positioning part (20A, 20B),

characterized in that

the reference surface (200A, 200B) includes two contact parts (P1, P2; P3, P4) provided apart from each other by a predetermined distance and restricting a rotational direction of the flange (130, 130A) as a result of

a part of the outer peripheral surface of the flange (130, 130A) contacting the contact parts (P1, P2; P3, P4).

2. The optical connector (10) according to claim 1,

wherein the reference surface (200A, 200B) includes a recessed part (210A, 210B), and

wherein the recessed part (210A, 210B) includes a bottom portion provided at a region interposed between the contact parts (P1, P2; P3, P4), the bottom portion facing the outer peripheral surface of the flange (130, 130A) without contacting the outer peripheral surface of the flange (130, 130A).

3. The optical connector (10) according to claim 2,

wherein the flange (130, 130A) includes a projection (133), and

wherein the projection (133) is provided on the outer peripheral surface of the flange (130, 130A) and is accommodated in the recessed part (210A, 210B).

4. The optical connector (10) according to any one of claims 1 to 3,

wherein the optical connector (10) has a floating structure, and

wherein the floating structure is such that the ferrule (110) is pushed toward the rear end portion of the optical connector (10) from the front end portion of the optical connector (10) to thereby vary an installation position of the ferrule assembly (100) in the housing (20, 30).

5. The optical connector (10) according to any one of claims 1 to 4,

wherein an external shape of the flange (130, 130A) is a quadrilateral shape when the flange (130, 130A) is viewed toward the rear end portion of the optical connector (10) from the front end portion of the optical connector (10).

6. The optical connector (10) according to claim 5,

wherein a pair of corners of the external shape are curved, at least one of facing sides of the external shape being interposed between the pair of corners.

7. The optical connector (10) according to any one of claims 1 to 6,

wherein the optical fiber (50) is any one of a multi-core fiber, a polarization maintaining optical fiber, and a bundle fiber.

8. The optical connector (10) according to any one of claims 1 to 7,

wherein the contact parts (P1, P2; P3, P4) are both regions having areas large enough to allow the flange (130, 130A) to be stably installed.